

Synoptic CB: Macrophyte Fluxes

2023 Samples

2026-03-16

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#Read in and collate all the files in the Processed Data folder

##Pull in Hydrago probe data

Some flux IDs are missing from hydrago data

```
## [1] "GCW_WC_1_202310" "GCW_WC_2_202310" "GCW_WC_3_202310" "GCW_WC_4_202306"
## [5] "GCW_WC_4_202307" "GCW_WC_4_202310" "GCW_WC_5_202310" "GWI_TR_1_202304"
## [9] "GWI_TR_1_202305" "GWI_TR_1_202307" "GWI_TR_1_202310" "GWI_TR_1_202311"
## [13] "GWI_TR_2_202304" "GWI_TR_2_202305" "GWI_TR_2_202307" "GWI_TR_2_202310"
## [17] "GWI_TR_2_202311" "GWI_TR_3_202304" "GWI_TR_3_202305" "GWI_TR_3_202307"
## [21] "GWI_TR_3_202310" "GWI_TR_3_202311" "GWI_TR_4_202304" "GWI_TR_4_202305"
## [25] "GWI_TR_4_202307" "GWI_TR_4_202310" "GWI_TR_4_202311" "GWI_TR_5_202304"
## [29] "GWI_TR_5_202305" "GWI_TR_5_202307" "GWI_TR_5_202310" "GWI_TR_5_202311"
## [33] "GWI_WC_1_202307" "GWI_WC_2_202307" "GWI_WC_3_202307" "GWI_WC_4_202307"
## [37] "GWI_WC_5_202307" "MSM_WC_1_202304" "MSM_WC_2_202304" "MSM_WC_2_202306"
## [41] "MSM_WC_3_202304" "MSM_WC_4_202304" "MSM_WC_5_202304" "SWH_WC_1_202307"
## [45] "SWH_WC_1_202308" "SWH_WC_1_202309" "SWH_WC_2_202306" "SWH_WC_2_202307"
## [49] "SWH_WC_2_202308" "SWH_WC_2_202309" "SWH_WC_2_202310" "SWH_WC_3_202306"
## [53] "SWH_WC_3_202307" "SWH_WC_3_202308" "SWH_WC_4_202306" "SWH_WC_4_202307"
## [57] "SWH_WC_4_202308" "SWH_WC_4_202309" "SWH_WC_5_202306" "SWH_WC_5_202308"
## [61] "SWH_WC_5_202309"
```

##Remove bad fluxes and calculate how many there were

Number flux measurements removed: 10

Number negative CO2 fluxes removed: 5

Number unusually high CH4 fluxes removed: 2

Number messy fluxes removed: 3

Percentage flux measurements removed: 5.24%

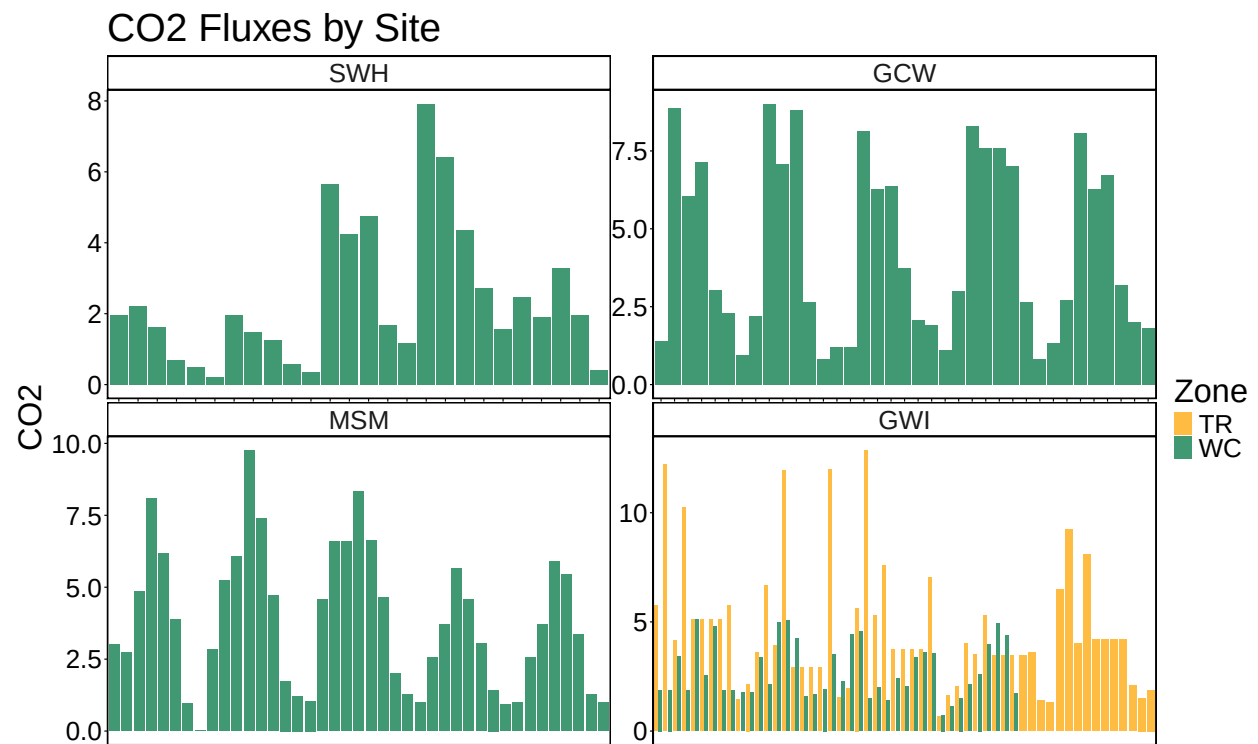
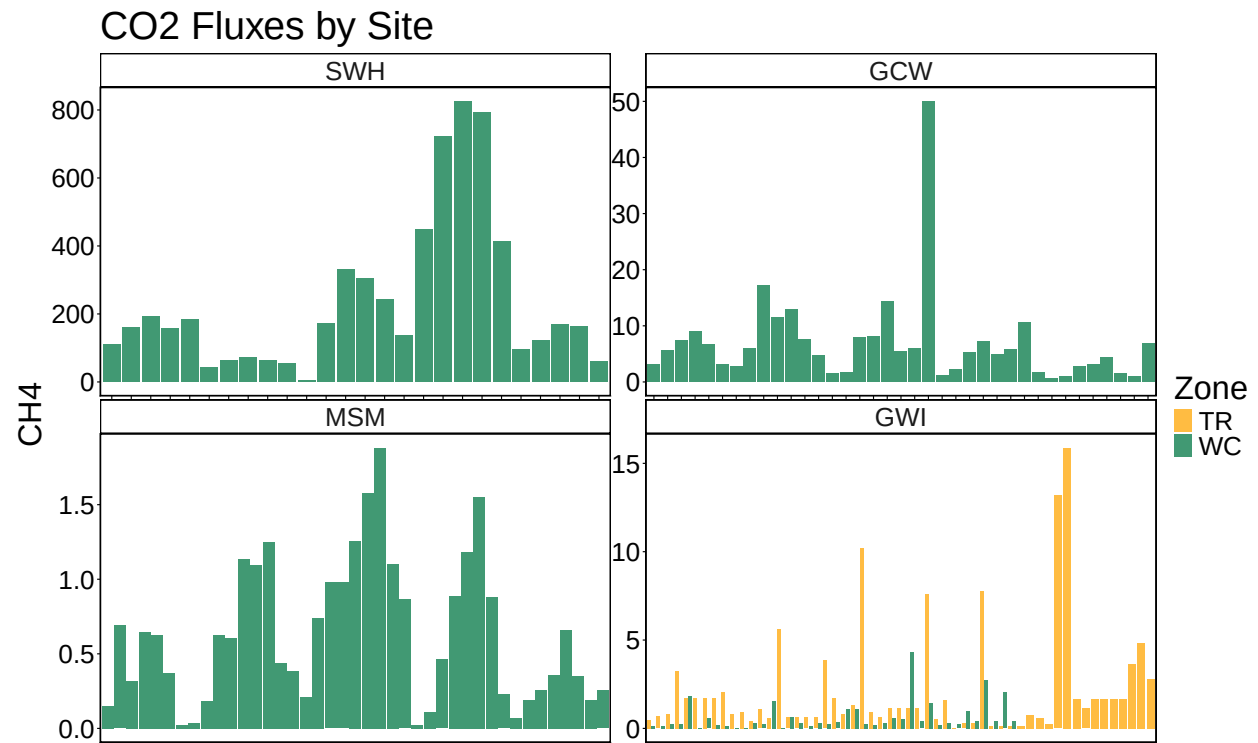
Percentage negative CO₂ fluxes removed: 2.62%

Percentage unusually high CH₄ fluxes removed: 1.05%

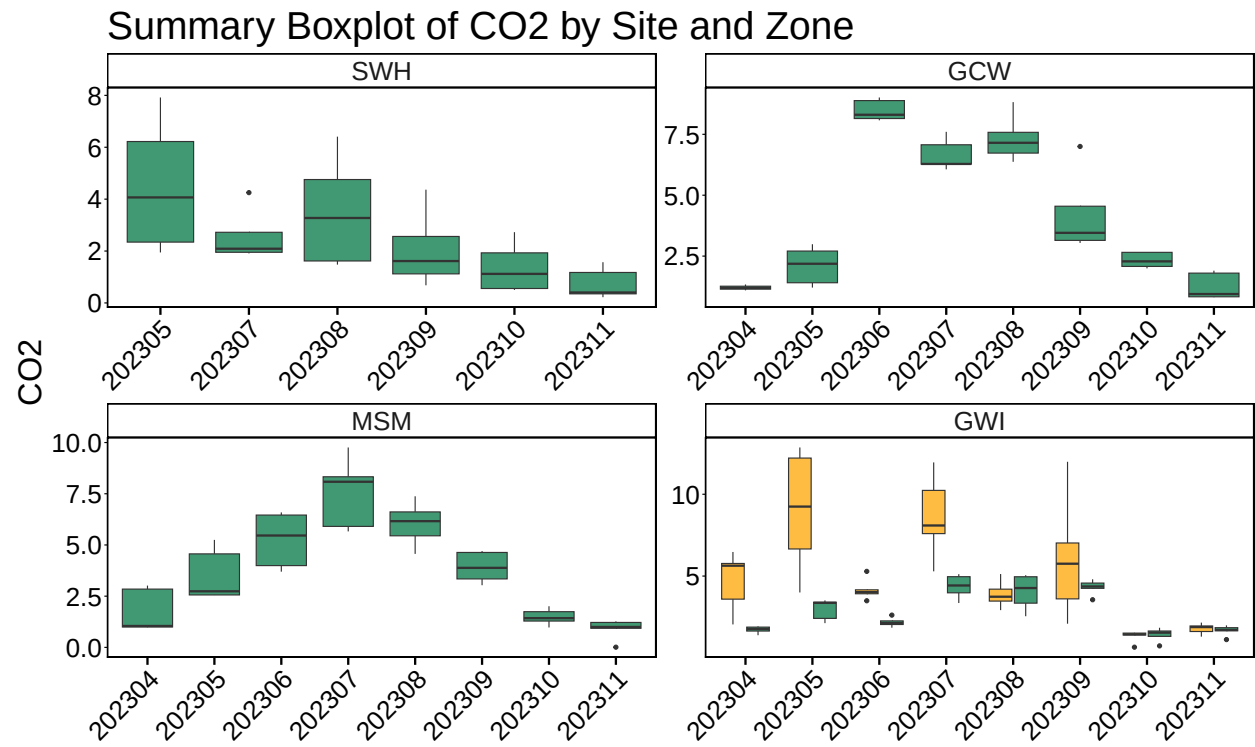
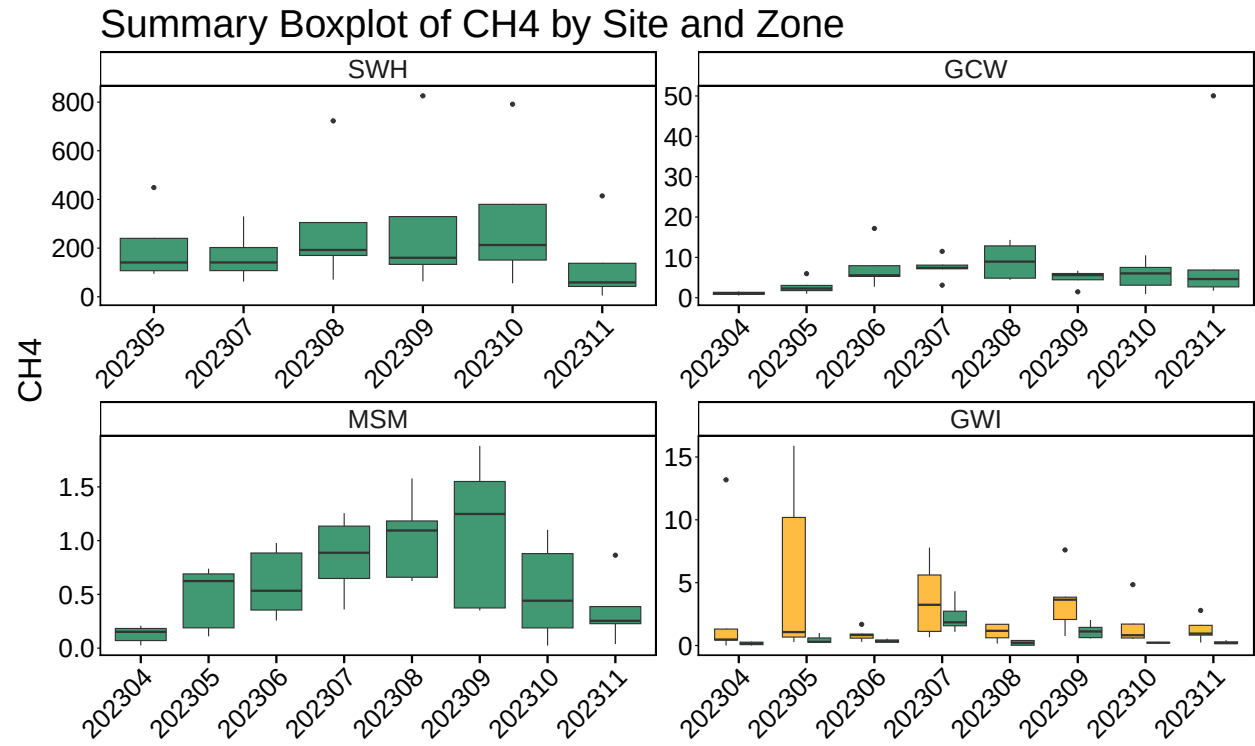
Percentage messy fluxes removed: 1.57%

0.1 Write out files

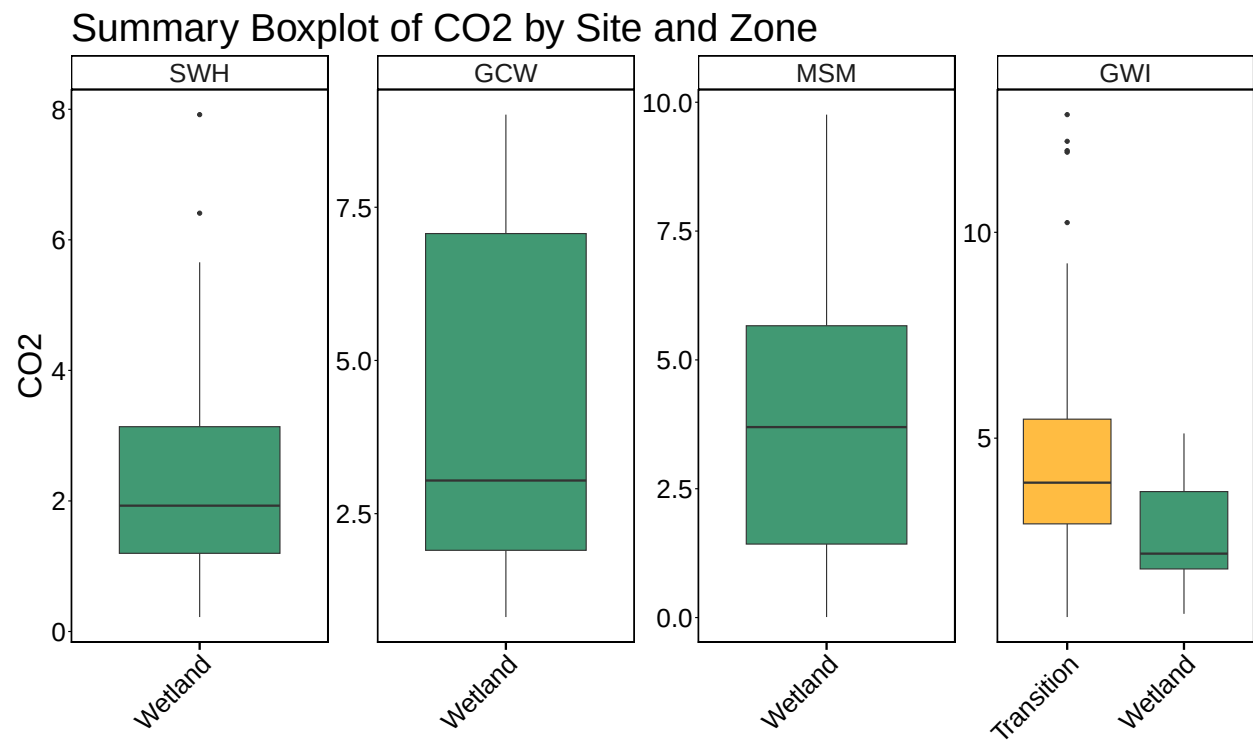
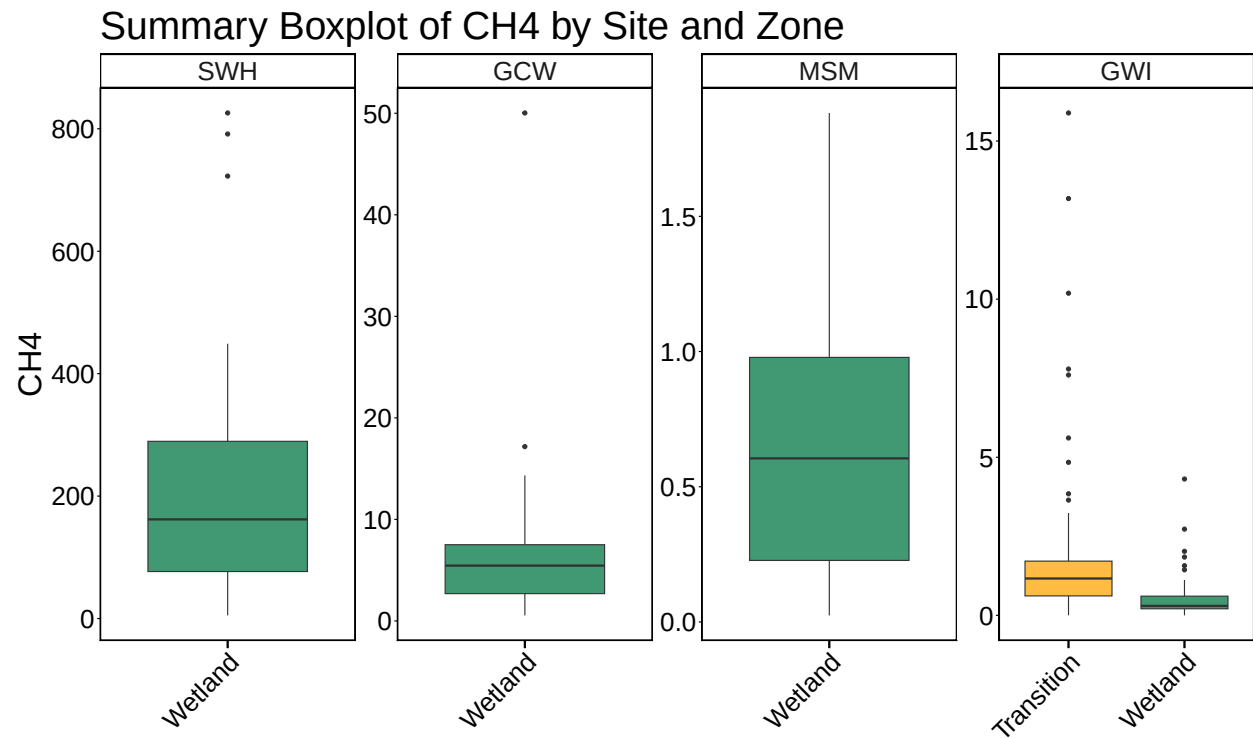
0.2 All Data by Year, Site and Zone



0.3 Boxplot summary by Date



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Wetland	5.1325387	825.834316	234.7312125	0.2214903	7.917877	2.435934
GCW	Wetland	0.5469638	50.029432	6.8147698	0.8129533	9.011439	4.361891
MSM	Wetland	0.0242469	1.882217	0.6297623	0.0101574	9.759568	3.839438
GWI	Transition	0.0020981	15.888051	2.1885762	0.6548599	12.858602	4.658330
GWI	Wetland	-0.0009012	4.312324	0.6393703	0.7326644	5.114817	2.782669